STA_445_Assignment 7

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Problem 1:

The `infmort` data set from the package `faraway` gives the infant mortality rate for a variety of countries. The information is relatively out of date, but will be fun to graph. Visualize the data using by creating scatter plots of mortality vs income while faceting using `region` and setting color by `oil` export status. Utilize a $\log_{10}$ transformation for both `mortality` and `income` axes. This can be done either by doing the transformation inside the `aes()` command or by utilizing the `scale_x_log10()` or `scale_y_log10()` layers. The critical difference is if the scales are on the original vs log transformed scale. Experiment with both and see which you prefer.

a. The `rownames()` of the table gives the country names and you should create a new column that contains the country names. *`rownames`

```
country_column <- infmort %>%
  mutate(country = rownames(infmort))

country_column %>%
  head(7)
```

```
##                        region income mortality            oil
## Australia                Asia   3426      26.7 no oil exports
## Austria                 Europe   3350      23.7 no oil exports
## Belgium                 Europe   3346      17.0 no oil exports
## Canada                Americas   4751      16.8 no oil exports
## Denmark                 Europe   5029      13.5 no oil exports
## Finland                 Europe   3312      10.1 no oil exports
## France                  Europe   3403      12.9 no oil exports
##                                   country
## Australia             Australia
## Austria               Austria
## Belgium               Belgium
## Canada                Canada
## Denmark               Denmark
## Finland               Finland
## France                France
```

b. Create scatter plots with the `log10()` transformation inside the `aes()`command.

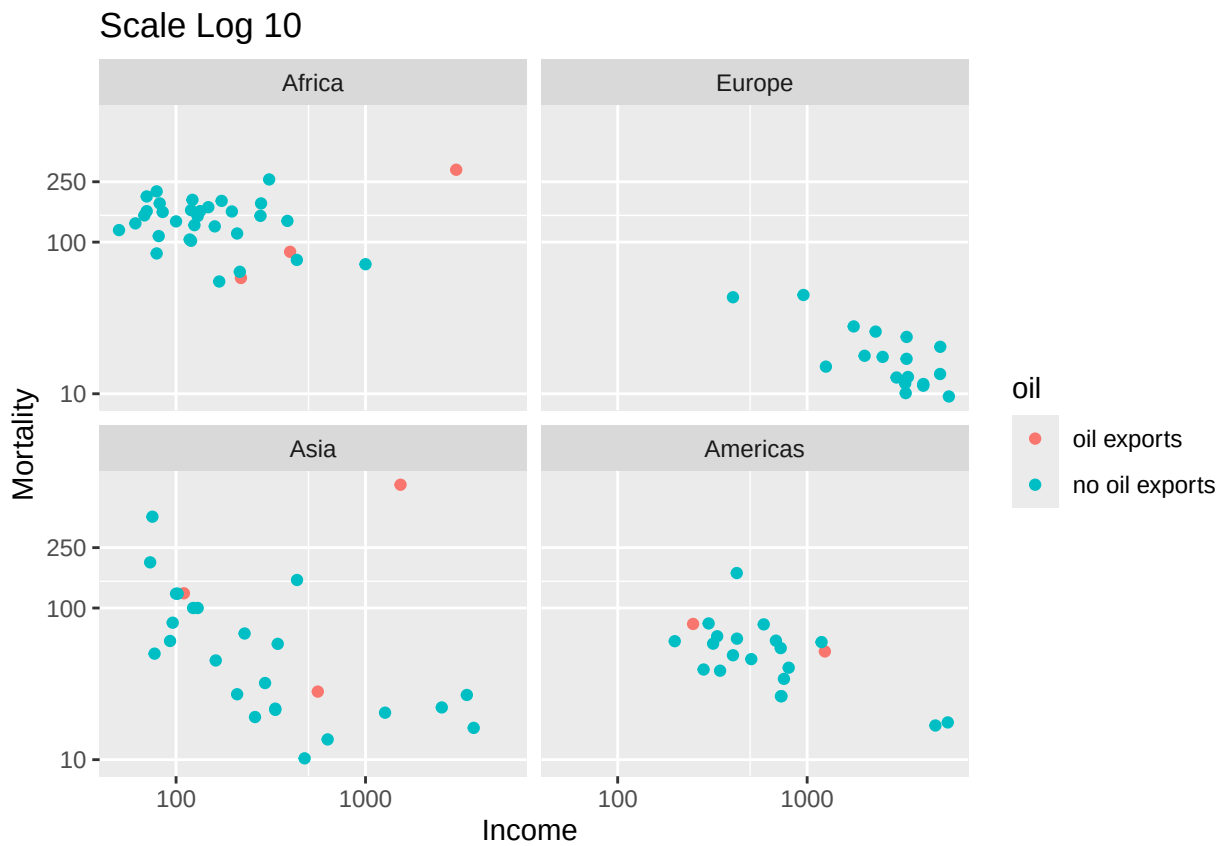
```r
ggplot(country_column, aes(x = log10(income), y = log10(mortality), color = oil)) +
  geom_point() +
  facet_wrap(~region) +
  labs(title = "Log10 mortality vs income",
       x = "log10 Income",
       y = "log10 Mortality")
```

```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

c. Create the scatter plots using the `scale_x_log10()` and `scale_y_log10()`.Set the major and minor breaks to be useful and aesthetically pleasing.Comment on which version you find easier to read.

```r
ggplot(country_column, aes(x = income, y = mortality, color = oil)) +
  geom_point() +
  scale_x_log10(breaks = c(10, 100, 1000),
                minor_breaks = c(0, 500)) +
  scale_y_log10(breaks = c(10, 100, 250),
                minor_breaks = c(0, 150, 250)) +
  facet_wrap(~region) +
  labs(title = "Scale Log 10",
       x = "Income ",
       y = "Mortality ")
```

```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



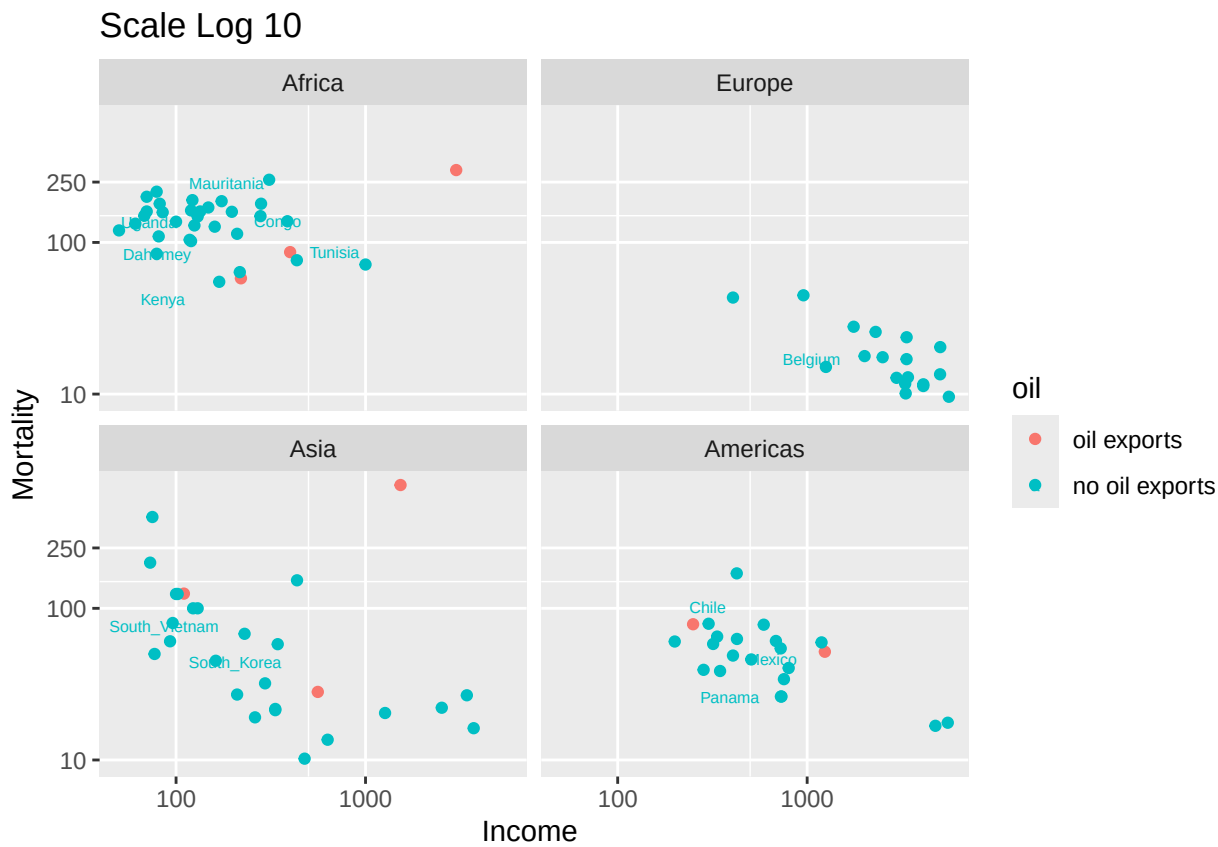
d. The package `ggrepel` contains functions `geom_text_repel()` and `geom_label_repel()` that mimic the basic `geom_text()` and `geom_label()`functions in `ggplot2`, but work to make sure the labels don't overlap. Select 10-15 countries to label and do so using the `geom_text_repel()` function.

```
sample <- country_column %>%
  slice_sample(n = 12)
```

```
ggplot(country_column, aes(x = income, y = mortality, color = oil)) +
  geom_point() +
  scale_x_log10(breaks = c(10, 100, 1000),
                minor_breaks = c(0, 500)) +
  scale_y_log10(breaks = c(10, 100, 250),
                minor_breaks = c(0, 150, 250)) +
  facet_wrap(~region) +
  geom_text_repel(data = sample,
                  aes(label = country),
                  size = 2) +
  labs(title = "Scale Log 10",
       x = "Income ",
       y = "Mortality ")
```

```
## Warning: Removed 4 rows containing missing values or values outside the scale range
```

```
## (`geom_point()`).
```



Problem 2

Using the `datasets::trees` data, complete the following:

a. Create a regression model for $y =$ `Volume` as a function of $x =$ `Height`.

```
(tree1 <- lm(Volume ~ Height, data = trees))
```

```
##
## Call:
## lm(formula = Volume ~ Height, data = trees)
##
## Coefficients:
## (Intercept)       Height
##     -87.124        1.543
```

b. Using the str(your model's name) command, to get a list of all the information stored in the linear model object. Use $ to extract the slope and intercept of the regression line (the coefficients).

```r
str(tree1)
```

```
## List of 12
##  $ coefficients : Named num [1:2] -87.12 1.54
##   ..- attr(*, "names")= chr [1:2] "(Intercept)" "Height"
##  $ residuals    : Named num [1:31] -10.6109 -2.8941 0.0926 -7.5976 -19.0877 ...
##   ..- attr(*, "names")= chr [1:31] "1" "2" "3" "4" ...
##  $ effects      : Named num [1:31] -167.98 53.86 2.05 -5.89 -17.64 ...
##   ..- attr(*, "names")= chr [1:31] "(Intercept)" "Height" "" "" ...
##  $ rank         : int 2
##  $ fitted.values: Named num [1:31] 20.9 13.2 10.1 24 37.9 ...
##   ..- attr(*, "names")= chr [1:31] "1" "2" "3" "4" ...
##  $ assign       : int [1:2] 0 1
##  $ qr           :List of 5
##   ..$ qr   : num [1:31, 1:2] -5.57 0.18 0.18 0.18 0.18 ...
##   .. ..- attr(*, "dimnames")=List of 2
##   .. .. ..$ : chr [1:31] "1" "2" "3" "4" ...
##   .. .. ..$ : chr [1:2] "(Intercept)" "Height"
##   .. ..- attr(*, "assign")= int [1:2] 0 1
##   ..$ qraux: num [1:2] 1.18 1.29
##   ..$ pivot: int [1:2] 1 2
##   ..$ tol  : num 1e-07
##   ..$ rank : int 2
##   ..- attr(*, "class")= chr "qr"
##  $ df.residual  : int 29
##  $ xlevels      : Named list()
##  $ call         : language lm(formula = Volume ~ Height, data = trees)
##  $ terms        :Classes 'terms', 'formula'  language Volume ~ Height
##   .. ..- attr(*, "variables")= language list(Volume, Height)
##   .. ..- attr(*, "factors")= int [1:2, 1] 0 1
##   .. .. ..- attr(*, "dimnames")=List of 2
##   .. .. .. ..$ : chr [1:2] "Volume" "Height"
##   .. .. .. ..$ : chr "Height"
##   .. ..- attr(*, "term.labels")= chr "Height"
##   .. ..- attr(*, "order")= int 1
##   .. ..- attr(*, "intercept")= int 1
##   .. ..- attr(*, "response")= int 1
##   .. ..- attr(*, ".Environment")=<environment: R_GlobalEnv>
##   .. ..- attr(*, "predvars")= language list(Volume, Height)
##   .. ..- attr(*, "dataClasses")= Named chr [1:2] "numeric" "numeric"
##   .. .. ..- attr(*, "names")= chr [1:2] "Volume" "Height"
##  $ model        :'data.frame':   31 obs. of  2 variables:
##   ..$ Volume: num [1:31] 10.3 10.3 10.2 16.4 18.8 19.7 15.6 18.2 22.6 19.9 ...
##   ..$ Height: num [1:31] 70 65 63 72 81 83 66 75 80 75 ...
##   ..- attr(*, "terms")=Classes 'terms', 'formula'  language Volume ~ Height
##   .. .. ..- attr(*, "variables")= language list(Volume, Height)
##   .. .. ..- attr(*, "factors")= int [1:2, 1] 0 1
##   .. .. .. ..- attr(*, "dimnames")=List of 2
##   .. .. .. .. ..$ : chr [1:2] "Volume" "Height"
##   .. .. .. .. ..$ : chr "Height"
##   .. .. ..- attr(*, "term.labels")= chr "Height"
##   .. .. ..- attr(*, "order")= int 1
##   .. .. ..- attr(*, "intercept")= int 1
```

```
##   .. .. ..- attr(*, "response")= int 1
##   .. .. ..- attr(*, ".Environment")=<environment: R_GlobalEnv>
##   .. .. ..- attr(*, "predvars")= language list(Volume, Height)
##   .. .. ..- attr(*, "dataClasses")= Named chr [1:2] "numeric" "numeric"
##   .. .. .. ..- attr(*, "names")= chr [1:2] "Volume" "Height"
##  - attr(*, "class")= chr "lm"
```

```
tree1$coefficients[1]
```

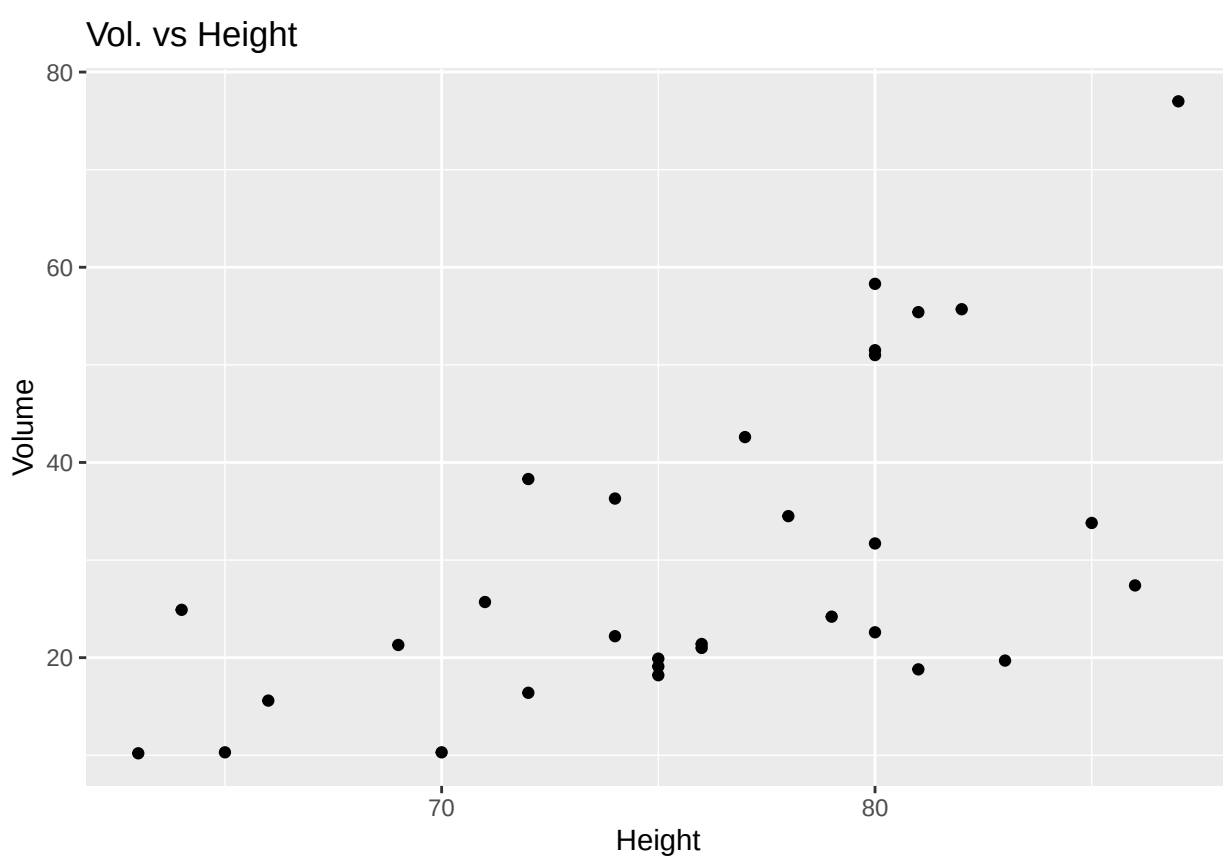
```
## (Intercept)
##   -87.12361
```

```
tree1$coefficients[2]
```

```
##  Height
## 1.54335
```

c. Using `ggplot2`, create a scatter plot of Volume vs Height.

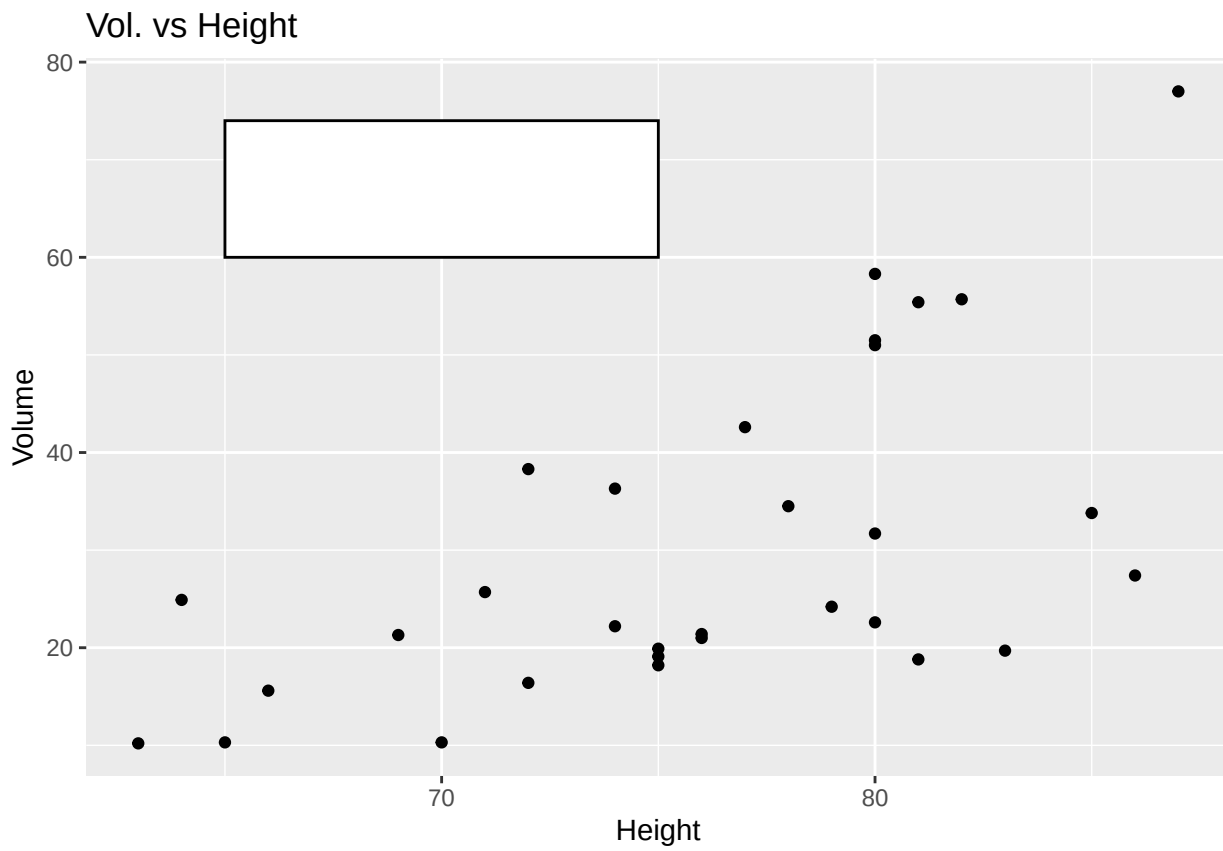
```
ggplot(data=trees, aes(x = Height, y = Volume )) +
  geom_point() +
  labs(title = "Vol. vs Height",
       x= "Height",
       y = "Volume")
```



d. Create a nice white filled rectangle to add text information to using by adding the following annotation layer.

annotate('rect', xmin=65, xmax=75, ymin=60, ymax=74, fill='white', color='black')

```
ggplot(data=trees, aes(x = Height, y = Volume )) +
  geom_point() +
  labs(title = "Vol. vs Height",
       x= "Height",
       y = "Volume") +
  annotate('rect', xmin=65, xmax=75, ymin=60, ymax=74, fill='white',
           color='black')
```



e. Add some annotation text to write the equation of the line $\hat{y}_i = -87.12 + 1.54x_i$ in the text area.

```
#ggplot(data=trees, aes(x = Height, y = Volume )) +
  #geom_point() +
  #labs(title = "Vol. vs Height",
       #x= "Height",
       #y = "Volume") +
  #annotate('rect', xmin=65, xmax=75, ymin=60, ymax=74, fill='white',
           #color='black') +
  #annotate('text', x=68, y=65,
           #color = 'black', size = 5,
           #label = TeX("$\\hat{y}_i = -87.12 + 1.54x_i$",
                      #output = "character"), parse = TRUE)
```

f. Add annotation to add $R^2 = 0.358$

g. Add the regression line in red. The most convenient layer function to use is `geom_abline()`.

Problem 3

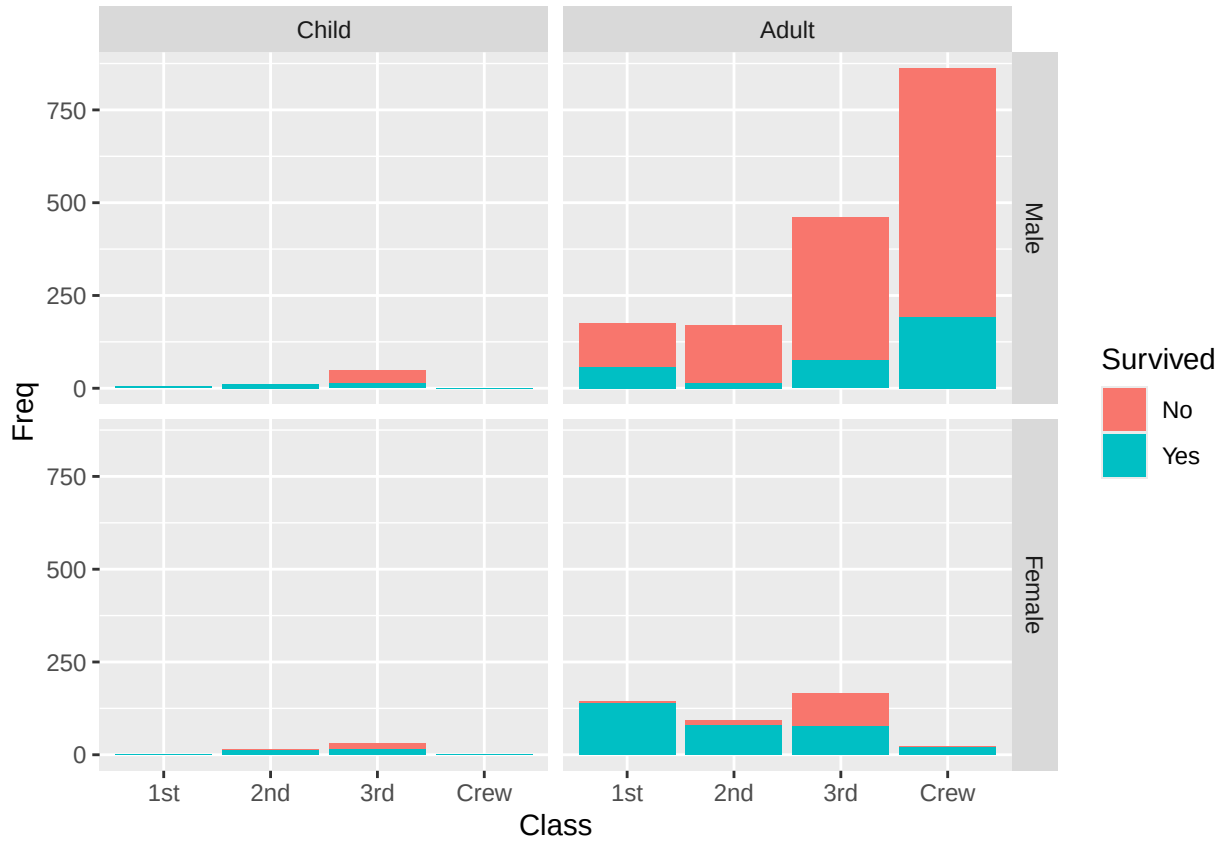
In `datasets::Titanic` table summarizes the survival of passengers aboard the ocean liner *Titanic*. It includes information about passenger class, sex, and age (adult or child). Create a bar graph showing the number of individuals that survived based on the passenger `Class`, `Sex`, and `Age` variable information. You'll need to use faceting and/or color to get all four variables on the same graph. Make sure that differences in survival among different classes of children are perceivable. *Unfortunately, the data is stored as a* `table` *and to expand it to a data frame, the following code can be used.*

```
‘‘‘ r
Titanic <- Titanic %>% as.data.frame()
print(Titanic)
‘‘‘
```

```
‘‘‘
##    Class    Sex   Age Survived Freq
## 1    1st   Male Child       No    0
## 2    2nd   Male Child       No    0
## 3    3rd   Male Child       No   35
## 4   Crew   Male Child       No    0
## 5    1st Female Child       No    0
## 6    2nd Female Child       No    0
## 7    3rd Female Child       No   17
## 8   Crew Female Child       No    0
## 9    1st   Male Adult       No  118
## 10   2nd   Male Adult       No  154
## 11   3rd   Male Adult       No  387
## 12  Crew   Male Adult       No  670
## 13   1st Female Adult       No    4
## 14   2nd Female Adult       No   13
## 15   3rd Female Adult       No   89
## 16  Crew Female Adult       No    3
## 17   1st   Male Child      Yes    5
## 18   2nd   Male Child      Yes   11
## 19   3rd   Male Child      Yes   13
## 20  Crew   Male Child      Yes    0
## 21   1st Female Child      Yes    1
## 22   2nd Female Child      Yes   13
## 23   3rd Female Child      Yes   14
## 24  Crew Female Child      Yes    0
## 25   1st   Male Adult      Yes   57
## 26   2nd   Male Adult      Yes   14
## 27   3rd   Male Adult      Yes   75
## 28  Crew   Male Adult      Yes  192
## 29   1st Female Adult      Yes  140
## 30   2nd Female Adult      Yes   80
## 31   3rd Female Adult      Yes   76
## 32  Crew Female Adult      Yes   20
‘‘‘
```

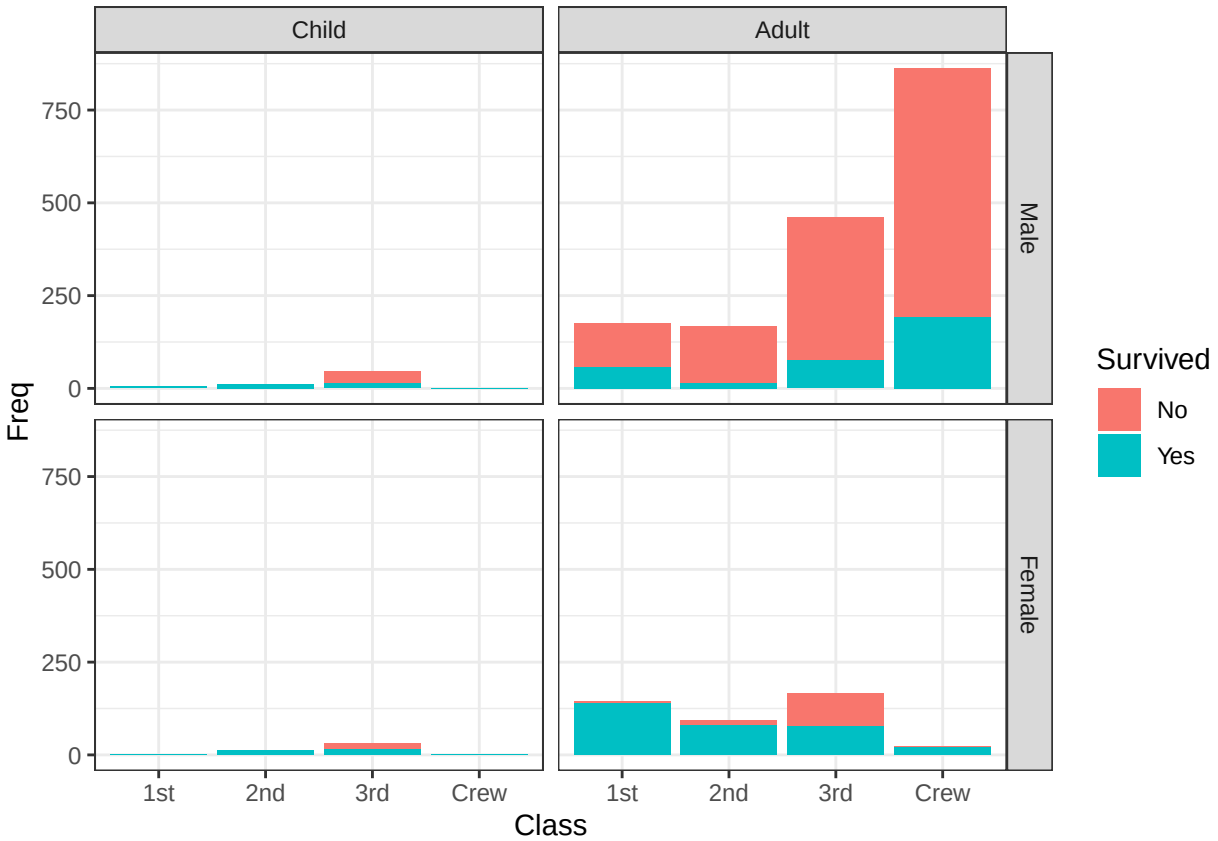
a. Make this graph using the default theme. *If you use color to denote survivorship, modify the color scheme so that a cold color denotes death.*

```
ggplot(data = Titanic, aes(x = Class, y = Freq, fill = Survived )) +
  geom_bar(stat = 'identity')+
  facet_grid(Sex~Age)
```



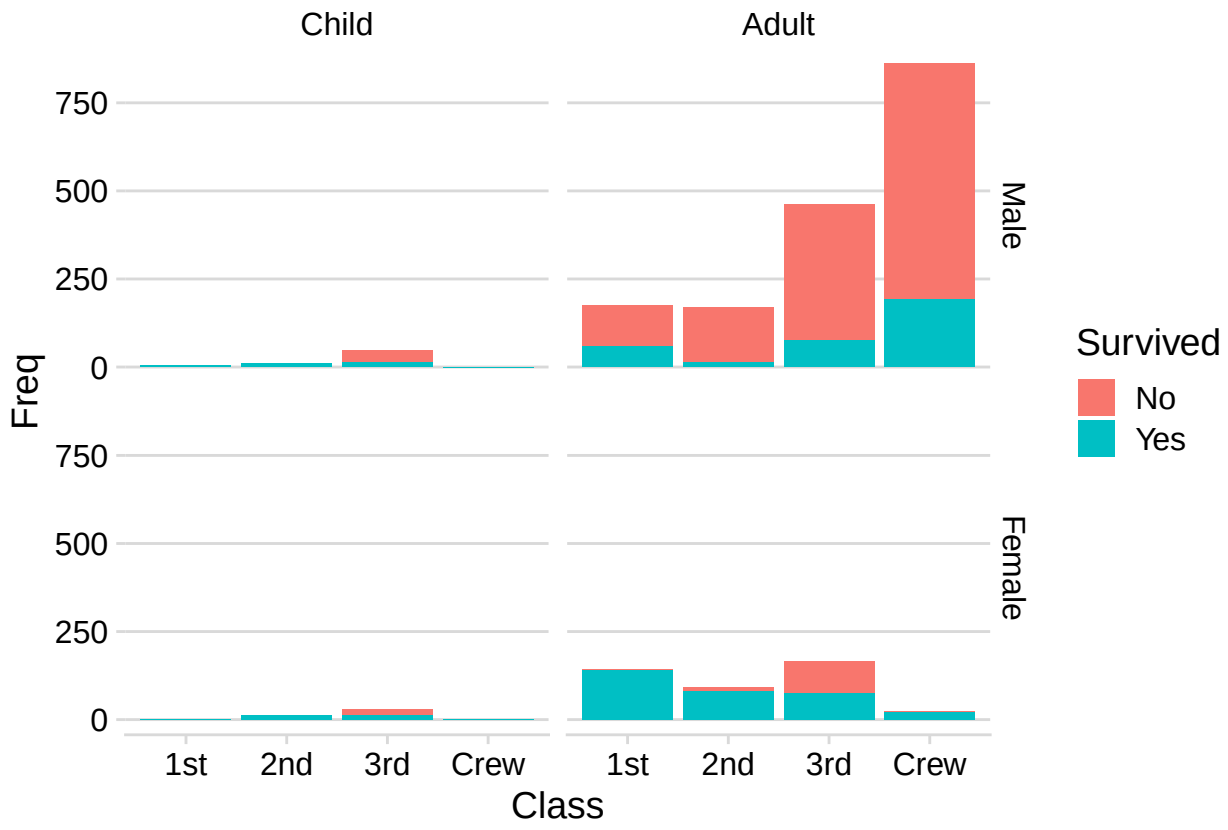
b. Make this graph using the `theme_bw()` theme.

```
ggplot(data = Titanic, aes(x = Class, y = Freq, fill = Survived )) +
  theme_bw() +
  geom_bar(stat = 'identity')+
  facet_grid(Sex~Age)
```



c. Make this graph using the `cowplot::theme_minimal_hgrid()` theme.

```
ggplot(data = Titanic, aes(x = Class, y = Freq, fill = Survived )) +
  geom_bar(stat = 'identity')+
  facet_grid(Sex~Age)+
  cowplot::theme_minimal_hgrid()
```



d. Why would it be beneficial to drop the vertical grid lines?

It makes the x-axis a little more clear and identifiable.